

Munachiso Samuel (Sam) Nwadike

Research Engineer (II) • Degrees: NYU, MBZUAI
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A.I. INTERESTS

I investigate the physics of intelligence, designing interpretable models that exhibit concurrent growth in sophistication and human alignment.

EDUCATION

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE

Master of Science, Machine Learning (4.0 GPA) January 2021 – December 2022

Thesis: Bayesian Optimisation for Hyperparameter Tuning

Advisor: Dr. Qirong Ho (University President's Center for Integrative Artificial Intelligence)

Key Courses: Advanced Machine Learning, Statistical Inference, Optimisation

Graduate Teaching Assistant, ML710, Parallel and Distributed Systems Course

New York University, Abu Dhabi, UAE

Bachelor of Science, Computer Science with Mathematics August 2015 – May 2019

Thesis: Semantic Segmentation of Satellite Imagery

Advisor: Dr. Yi Fang (Multimedia and Visual Computing Lab)

Key Courses: Real Analysis, Measure Theory & Advanced Probability, Differential Equations, Machine Learning, Computer Vision

Whitesands School, Lagos, Nigeria

High School Valedictorian

June 2014

Best in Mathematics, Physics, Chemistry, Biology, IT, English.

School-record GCSE score

MEDIA & PRESS

- *Khaleej Times*
 - ❖ "New UAE AI turns raw data into business insights ..." (27/09/2025)
<https://www.khaleejtimes.com/uae/ai-turns-raw-data-into-business-insights> (please open on mobile for better quality photos)
- *MBZUAI News*.
 - ❖ "How a team of researchers from MBZUAI is using AI to empower businesses with instant data analytics" › <http://bit.ly/4nY71zG> (30/09/2025)
 - ❖ "Overcoming the 'reversal curse' in LLMs with ReCall" › <http://bit.ly/4ORstgT> (28/07/2025)

AWARDS & ACHIEVEMENTS

ICML Gold Reviewer Award, (top machine learning conference), 2026

UAE Golden Visa for Talented Persons/Specialists in Science, 2023

Semifinalist Team Leader, MBZIRC Robotics Competitions, 2022

MBZUAI Fellowship Award (4.5% acceptance rate), 2021

USD \$8000 Deep Learning Indaba IndabaX *Innovation Grant* Awardee, 2020

Winner, CEO for a Day, Refinitiv Data Co. (London Stock Exchange), 2019

NYU Scholarship Award (~3.7% acceptance rate), 2016

**PUBLICATIONS
& PREPRINTS**

1. (Preprint, 2026) "Einstein World Models." Nwadike, M. S., Iklassov, Z., Mekky, A., Zuhri, Z., & Inui, K. arXiv:2606.26969.
2. (Preprint, 2026) "Measuring AI Reasoning: A Guide for Researchers." Nwadike, M. S., Iklassov, Z., Ali, K., Genadi, R., & Inui, K. arXiv:2605.02442.
3. (EACL 2026, Long Papers) "Sycophancy Hides Linearly in the Attention Heads." Genadi, R. A., Nwadike, M. S., Mukhituly, N., Hiraoka, T., AlQuabeh, H., & Inui, K. Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics, Rabat, Morocco, March 24–29.
4. (Preprint, 2025) "The AI Data Scientist." Akimov, F., Nwadike, M. S., Iklassov, Z., & Takáč, M. arXiv:2508.18113.
5. (WACV 2026, Main Conference) "MAFM³: Modular Adaptation of Foundation Models for Multi-Modal Medical AI." Qazi, M. A., Nwadike, M. S., Almakky, I., Yaqub, M., & Saeed, N. IEEE/CVF Winter Conference on Applications of Computer Vision, Tucson, Arizona, USA, March 6–10.
6. (ACL 2025, Main Conference) "RECALL: Library-Like Behavior in Language Models is Enhanced by Self-Referencing Causal Cycles." Nwadike, M. S., Iklassov, Z., Aremu, T., Hiraoka, T., Bojkovic, V., Heinzerling, B., AlQuabeh, H., Takáč, M., & Inui, K. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Vienna, Austria, July 27–August 1.
7. (Preprint, 2025) "Where Grokking Happens: Mechanistic Insights at the Embedding Layer." AlQuabeh, H., Bojkovic, V., Nwadike, M. S., & Inui, K. arXiv:2505.15624.
8. (Preprint, 2025) "Number Representations in LLMs: A Computational Parallel to Human Perception." AlQuabeh, H., Bojkovic, V., Nwadike, M. S., El-Shangiti, A. O., Hiraoka, T., & Inui, K. arXiv:2502.16147.
9. (ACII 2025, Main Conference) "Improving Personalisation in Valence and Arousal Prediction using Data Augmentation." Nwadike, M. S., Li, J., & Salam, H. 13th International Conference on Affective Computing and Intelligent Interaction, Canberra, Australia, October 8–11.
10. (Preprint, 2023) "Memoization-Aware Bayesian Optimization for AI Pipelines with Unknown Costs." Essofi, A., Salahuddeen, R., Nwadike, M. S., Kumar, N., et al. arXiv:2411.03731.
11. (AAAI Workshop, 2021) "Explainability Matters: Backdoor Attacks on Medical Imaging." Nwadike, M. S., Miyawaki, T., Sarkar, E., Maniatakos, M., & Shamout, F. Trustworthy AI for Healthcare Workshop, AAAI 2021, Virtual, February 9.

**PEER-REVIEW
& COMMUNITY**

Academic Peer-Reviewing & Program Committees

- Reviewer, ICML 2026 (6 papers)
- Reviewer, AAAI 2026 (2 papers)
- Reviewer, EMNLP 2025 (1 paper)

Community Participation

- *Delta Institute, AI Mentor (2024)*: Assisting and Mentoring High School students as they navigate future career paths relevant to AI
- *MBZUAI Student Careers Office (2021)*: Award University Community participation award for my as a student assistant to the inaugural Careers and Alumni Services

RESEARCH EXPERIENCE

- **Dual Affiliation: MBZUAI & Japan's Riken Center for Advanced Intelligence Project (July 2024 – Present)**
Mechanistic Interpretability: Working with Dr. Kentaro Inui on the fundamental mechanics of intelligent language systems. Responsibilities include: Mentoring MSc and PhD students, publishing top-tier research papers, communicating with marketing department of MBZUAI to translate research into articles for the public
- New York University Center for Artificial Intelligence and Robotics (June 2023 – June 2024)
Multimodal Affective Computing: Programmed systems for research in the automatic detection of human affect using deep learning, particularly for emotion prediction
- MBZUAI Center for Integrative Artificial Intelligence (Jan 2022 – Dec 2022)
Statistical Hyperparameter Tuning: Developed bayesian optimisation algorithms that allow scientists to tune AI hyperparameters despite them having stringent GPU and cost budget constraints
- MBZUAI Intelligent Visual Analytics Lab (May 2021 – Dec 2021)
Object Detection: Analysed the applicability of state of the art approaches to improving Meta Detectron 2 performance on open-world object detection, leveraging the latest insights from research on manifold mixup
- New York University Clinical Artificial Intelligence Lab (Feb 2020 – Oct 2020)
Security In AI for Healthcare: Innovated novel threat models for adversarial attacks in chest X-ray disease detection, and demonstrated mitigation methods using computer vision interpretability tools
- New York University Multimedia and Visual Computing Lab (Jan 2019 – Dec 2019)
3D Vision and Semantic Segmentation Research: Worked on semantic segmentation of satellite images, a project which we developed further into my undergraduate thesis. Also assisted with preprocessing tasks for 3D point cloud datasets

SKILLS

- Programming Languages: Python, C/C++, Matlab, LaTeX
- Machine Learning: BoTorch, PyTorch, Tensorflow, Keras, NumPy
- Cloud and Computing: vLLM, Docker, Bash, Kubernetes, MicroK8s, GCP, AWS, Slurm, Linux, Unix, Argo Workflows
- NLP: NLTK, SpaCy, ForteNLP, ChatGPT, Language Models
- Robotics: ROS2, Gazebo, Point Cloud Library (PCL)
- Languages: Advanced French, Intermediate Chinese (Mandarin)
- Interests: Public Speaking, Scientific Reading, Language Acquisition
- Organizational: PowerPoint, Word, Advanced Excel, Microsoft Project, Smartsheet, Notion

ARCHIVED PROJECTS

MBZIRC robotics competition (2022) [[whitepaper](#)]: Led the MBZIRC team to the semi-finals after writing our whitepaper for laser-guided drone swarms. Completed ROS2 and Gazebo simulation for SOTA sea-vessel detection in-simulation using computer vision

Phone-Based Sign Language Detection using AI (2019) [[report](#)]: Custom mobile app that allowed reading of american sign language (ASL) using solely a phone camera. We created the dataset inhouse and I was responsible for all AI and model training parts of the project

REFERENCES

1. Dr. Kentaro Inui: Professor, Language Model Interpretability Team, NLP Department, MBZUAI. Email: martin.takac@mbzuai.ac.ae
2. Dr. Martin Takáč: Associate Dean of Academic Affairs, Computing & Mathematical Sciences Division, MBZUAI. Email: martin.takac@mbzuai.ac.ae
3. Dr. Zangir Iklassov: Assistant Teaching Professor and Research Scientist at MBZUAI. Email: zangir.iklassov@mbzuai.ac.ae
4. Dr. Qirong Ho: CTO of Petuum, and Associate Professor of Machine Learning at MBZUAI. Email: qirong.ho@mbzuai.ac.ae
5. Dr. Jerome White: Professor of Computer Science at NYU Abu Dhabi and Researcher at Wadhvani AI. Email: jerome@wexfordheights.net